MOHAMMAD TALHA

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I am a CSE student with a passion for problem-solving and technology, eager to tackle dynamic challenges and apply analytical skills to real-world applications. I have demonstrated proficiency in data analysis, programming, and event management, along with strong collaboration and leadership abilities.

Area of Expertise

Data Analysis & Visualization	Event Management & Coordination	Web Development
Team Leadership	Communication & Documentation	Programming
Database Management	Microsoft Office Suite	Shift Coordination

Achievement

Champion – Hack for Humanity 2026
- Won Champion at Hack for Humanity, presented by AI Collective Bangladesh.

Champion – ULAB EEE Fest 2026
- Won the championship for an IoT-based Smart Safety Helmet.

ULAB CSE Project Competition 2026
- Secured 4th position among 55 projects with Notun Alo.

Certified Data Analyst
- Completed Data Analysis certification from DataSolution360 Institute.

Projects

Notun Alo - Smart AI Waste Management & Recycling Platform
- Built an AI-powered recycling platform featuring automated agency assignment using a multi-factor weighted scoring algorithm, Gradient Boosting-based pickup ETA prediction, and K-Means clustering for optimized service zones across Dhaka.
- Integrated YOLO-based plastic detection trained on a custom dataset of 300 images to identify plastic waste and automatically allocate reward points, alongside a Bengali-first Claude AI chatbot for natural language pickup scheduling, gamified rewards, badges, and leaderboards. Implemented a 3-tier fault-tolerant architecture with automatic ML failover.
- *Tech Stack: PHP, Python, Flask, MySQL, Scikit-learn, YOLO, JavaScript, HTML5, CSS, Claude API*

IoT-Based Smart Safety Helmet
- Developed an IoT-enabled smart helmet using ESP32 with alcohol, drowsiness, crash, and helmet-wearing detection for active rider safety.
- Integrated automated engine cutoff, buzzer alerts, and real-time safety monitoring using Blynk and Firebase.
- *Tech Stack: ESP32, Arduino, IoT, MPU6050, MQ-3, MAX30102, Blynk, Firebase*

Fish Recommendation System (AquaRec)
- Developed an AI-powered fish recommendation system using Random Forest & KNN with SMOTE, suggesting species based on water quality (pH, temperature, turbidity).
- Built a Streamlit web app for real-time predictions, improving aquaculture decision-making and yield.
- *Tech Stack: Python, Streamlit, Scikit-learn, Joblib, MySQL, HTML5, CSS*

Exam Management System
- Built a dual-portal (Admin/Student) system with secure question management, result tracking, login, profile updates, and instant exam results.
- *Tech Stack: Java, MySQL*

Volunteer Experience

Event Associate – Hult Prize ULAB 2026-27
Event Admin – Idea Hunters 5.0
Event Admin – MIND SHIFT-2025
Event Admin – Idea Hunters 4.0
Organizing Secretary – MISHAS
Campus Ambassador – Z-TechX
Campus Ambassador – Kaizen SUST (OmniStart 1.0)
Campus Ambassador – ICT Olympiad Bangladesh S2
Volunteer – ULAB Communication Office

Internship

ICT Olympiad Bangladesh S3 (Course Curriculum Team) *Sept 2025 – Jan 2026*

- Managed class hosting, recordings, and video editing for course delivery.
- Organized Q&A datasets, assisted in curriculum development, and coordinated with instructors.

Education

University of Liberal Arts Bangladesh *Sept 2022 – Present*

Bachelor in CSE

- Current CGPA more than 3.5.
- Successfully completed projects on Java, Python, MySQL, and Frontend.

Tamirul Millat Kamil Madrasah *2019 – 2021*

Arabic Medium

- Finished Higher School Certificate in 2021 with 5.00 GPA.
- Champion (twice) in cricket tournament.

Tamirul Millat Kamil Madrasah *2017 – 2019*

Arabic Medium

- Finished Senior School Certificate in 2019 with 5.00 GPA.

Additional Information

- **Languages:** Bengali (Native), English (Professional Fluency), Hindi.
- **Technical Skills:** Microsoft Suite (Word, Excel, PowerPoint, Power BI), Programming (HTML5, CSS, Java, SQL, Python), Video Editing (CapCut).